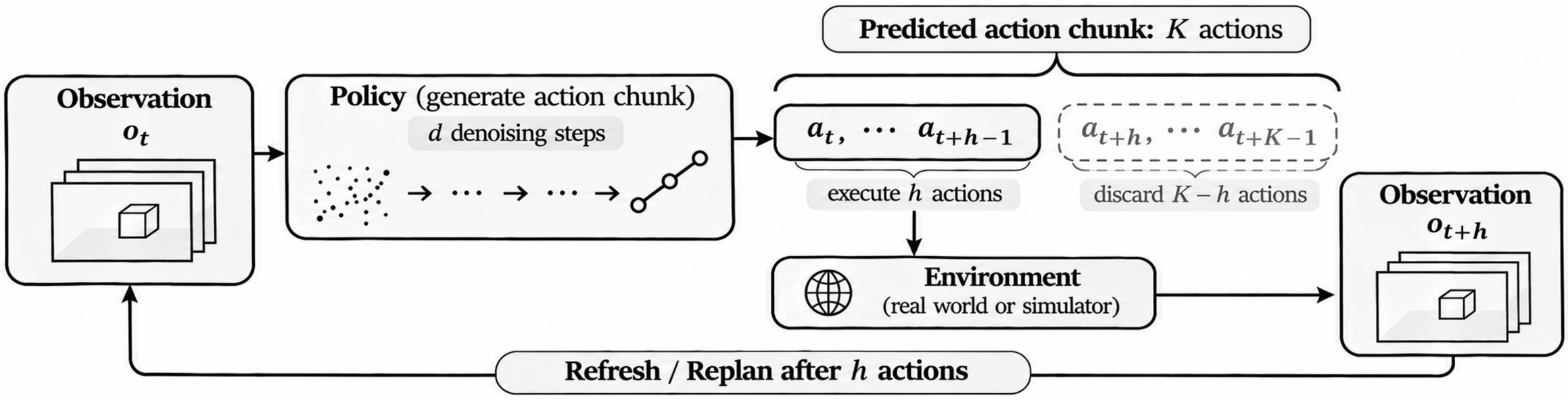
Refinement vs. Refresh

Compute Allocation for Closed-Loop Robot Policies

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Generation Effort vs. Execution Prefix



Parameters | K : number of predicted actions per chunk (chunk length) | h : executed actions per replan ($h \leq K$) | d : denoising steps (refinement)

Synchronous closed-loop execution. Each policy call predicts K actions; only the first h are executed before obtaining a new observation and replanning.

Evaluation Framework: Separate Design, Measurement, and Evidence Status

1. Define

Choose generation control r and execution prefix h.

2. Control

Fix the checkpoint and paired rollout conditions.

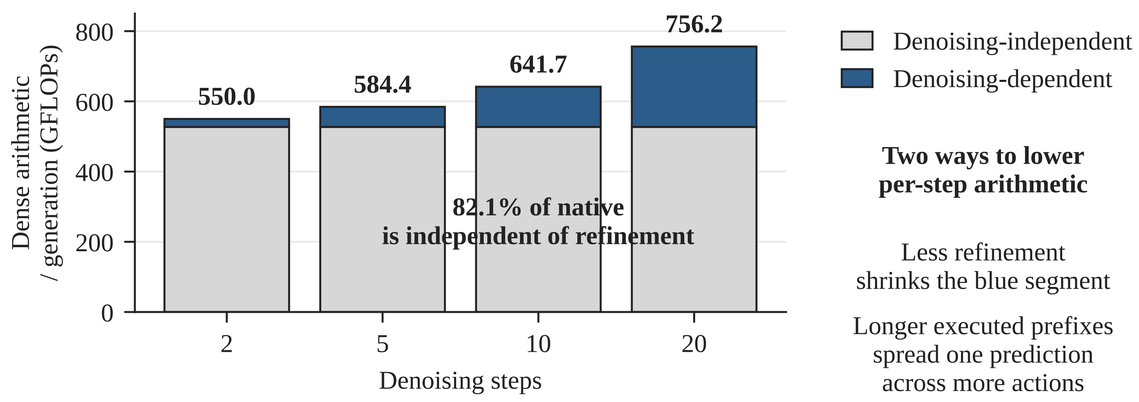
3. Account

Measure full generation and episode expenditure.

4. Freeze

Prespecify tests; label later contrasts descriptive.

SmolVLA: Complete Compute Accounting

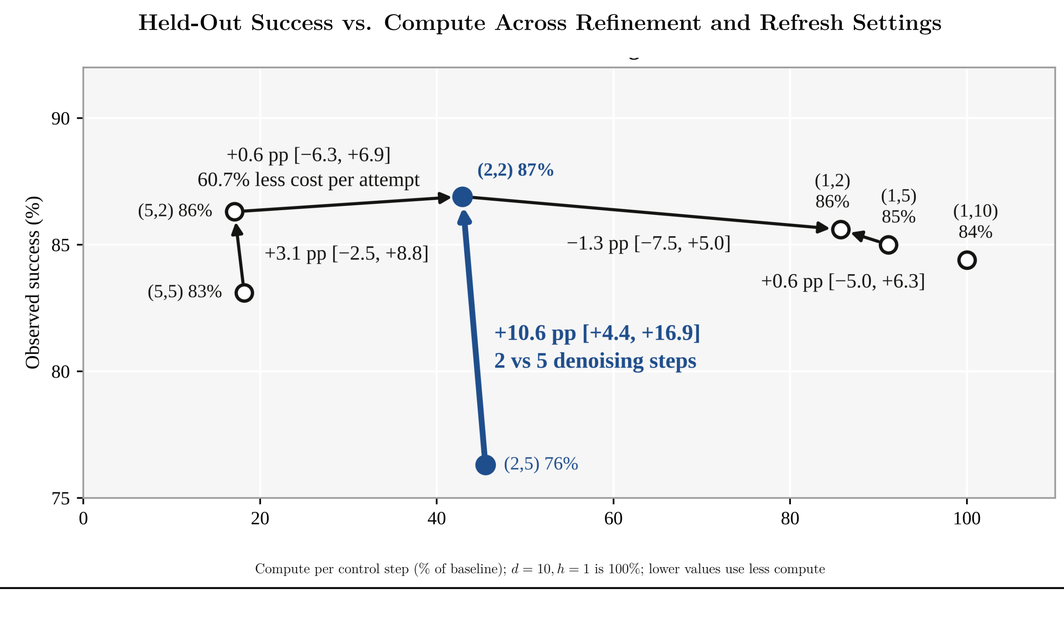


Two ways to lower per-step arithmetic

Less refinement shrinks the blue segment

Longer executed prefixes spread one prediction across more actions

Worked Example: Frozen SmolVLA



Evidence Status: What the SmolVLA Example Supports

Frozen before confirmation

At fixed refresh frequency, reducing denoising from five to two steps increased success from 122/160 to 139/160. This is the prespecified primary comparison.

Amended: descriptive only

Refreshing after five rather than two actions produced 138/160 versus 139/160 successes with 60.7% lower arithmetic per attempt. Similar observed success does not prove equal quality, and arithmetic is not deployment latency.

General recipe: define the policy's generation-effort control and execution prefix; measure success and complete expenditure together.
SmolVLA is one instantiation, not a cross-policy rule; keep prespecified and later descriptive evidence separate.